AR Interior Design App

T.E Mini Project work in partial fulfillment of the requirements

For the degree of

Bachelor of Technology

(Information Technology)

by

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Certificate

This is to certify that, the Mini-Project titled

"AR Interior Design App "

is a bonafide work done by

Divya Raghkumar Pichika(22IT1092)
Tanisha Indranil Maulik(22IT1094)

and is submitted in the partial fulfillment of the requirement for the degree of

Bachelor of Technology

(Information Technology)

to the

D.Y.Patil Deemed to be University.

Supervisor(Dr Jayanand Gawande)

Coordinator	Head of Department	Principal
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Mini Project Approval for T. E.

This is to certify that the Mini project entitled "**AR Interior Design App**" is a bonafide work done by **Divya Raghkumar Pichika**, **Tanisha Indranil Maulik** of V semester Information Technology, under the supervision of **Dr Jayanand Gawande**. This report has been approved.

Examiners:

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2: ______________________________

Supervisors:

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Declaration

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Date: ______________________

Abstract

This paper presents an augmented reality (AR) interior design application aimed at revolutionizing the way individuals conceptualize and personalize their living spaces. Leveraging the immersive capabilities of AR technology, the application allows users to visualize furniture and decor in real-time within their own homes, enabling them to make informed decisions before making purchases. Through a user-friendly interface and advanced rendering techniques, users can experiment with various designs, layouts, and styles, fostering creativity and confidence in their design choices. Furthermore, the app incorporates machine learning algorithms to provide personalized recommendations based on user preferences and existing room aesthetics. By combining the convenience of online shopping with the tangibility of physical space, this AR interior design app empowers users to create spaces that truly reflect their unique tastes and lifestyles

Keywords:
AR,interior design application,AR technology,user friendly.

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Abbreviations

AR	Augmented Reality
BIM	Building Information Modeling
VR	Virtual Reality
UI	User Interface
UX	User Experience
3D	Three-Dimensional
FF/E	Furniture, Fixtures, and Equipment
CMYK	Cyan, Magenta, Yellow, Key
RGB	Red, Green, Blue
XR	Extended Reality

Chapter 1

Introduction

1.1 Motivation

AR allows users to visualize how furniture, decor, and color schemes will look in their space before making any purchases or design decisions. This helps to prevent costly mistakes and ensures satisfaction with the final design.Clients can personalize their spaces by experimenting with different furniture layouts, styles, and colors in real-time.AR apps provide a convenient way for clients to explore design options without the need for physical prototypes or samples.

1.2 Objective

It enable users to visualize how furniture, decor, and other design elements will look and fit within their real-world spaces. AR allows users to overlay digital representations onto their physical environment, providing a more accurate and immersive preview of design choices.Assist users in optimizing the layout and functionality of their interior spaces. Through AR, users can virtually place and rearrange furniture to achieve optimal flow, functionality, and spatial organization.Ensure seamless integration with existing interior design workflows and processes.

1.3 Contribution

It focuses on improving the user experience by streamlining the interface, making it intuitive to navigate, and ensuring that AR features are seamless and easy to use.Research and add a wide range of furniture, decor items, and materials to the app's catalog, giving users plenty of options to visualize and experiment with in their spaces.Introduces customization options that allow users to adjust the size, color, and style of furniture and decor items to better match their preferences and existing interior design elements.Implement features that enable users to collaborate with others, such as sharing designs, commenting on designs, or even co-designing projects in real-time.Facilitate seamless purchasing by integrating the app with e-commerce platforms, allowing users to directly purchase items they like from within the app.Incorporate tools that enable users to accurately measure their rooms and plan layouts, ensuring that furniture and decor items fit well in their intended spaces.Provide curated collections of design inspiration,recommendations and trends to help users discover new ideas and styles for their interior design projects.Foster a sense of community among users by incorporating social features such as user profiles, forums, and the ability to follow other users and designers.It can also help the users to plan out their wallpaper and wall paints.

1.4 Problem Definition

The goal of this work is to make it user-friendly and provide unlimited customization options.Interior design is a complex and subjective process that often requires clients to visualize how different elements will look and feel within their space. Traditional methods of design presentation, such as mood boards and 2D renderings, can be limited in conveying the full potential of a design concept. Additionally, clients may struggle to imagine how furniture, decor, and color schemes will integrate into their existing environment.

1.5 Organization of Report

Chapter 2 deals with literature survey of AR interior design app. Chapter 3 contains proposed methodology and system framework of the application.Chapter 4 shows the system implementation.Chapter 5 contains the result and analysis of the project.Chapter 6 concludes the paper and mentions the future scopes of the project.

Chapter 2

Literature Survey

In the Interior Design Program, Emerging Technologies (AR, VR, and MR) A. S. Aalkhalidi, M. Izani, and A. A. Razak.

For many years, traditional presentation methods in interior design schools have made heavy use of sketching, 2D illustration, 3D graphics, animation, and drawings. The primary focus of this study is the potential use of cutting-edge technology in the interior design program in the United Arab Emirates (UAE). We are interested in the creative solutions that technologies such as mixed reality (MR), augmented reality (AR), and virtual reality (VR) can provide for interior design programs. This study looks at the applications, challenges, causes, impacts, and recommended solutions for interior design programs.

These technologies are widely used in a variety of industries, including tourism and healthcare, but there hasn't been much integration of them into interior design education and practice.

The method that was suggested was able to attain a 95 percentage accuracy rate for AR interior design applications and a 3.8 percent naturalness rate for speech synthesis.

Enhanced Visualization: Compared to typical 2D sketches or even 3D models, clients can visualize designs more precisely because to the immersive virtual environments that can be created by designers using AR, VR, and MR. [1]

S. Khan, K. Kadir, T. A. Khan, M. N. Zahid, and S. Nasir An application of augmented reality for interior designers and architects is Interno An economical resolution.

The software is meant to be used by architects and interior designers.This suggested research most likely acts as a useful instrument that can bridge the gap between industrial organizations and their clients, in addition to other pertinent company groups. It will facilitate visualization of architect plans and interior design concepts. A real setting can be designed virtually before it is physically installed. This allows architects and interior designers to view their 3D renderings on 2D drawings and virtually execute ideas in the chosen workspace before viewing them in real life.

Certain augmented reality apps might not be available to users without the necessary equipment because they might only be compatible with a certain kind of hardware or operating system.

For the AR interior design app, the suggested method obtained 94 percent accuracy, and for speech synthesis, it achieved 3.7 percent naturalness.

Accurate Tracking: Using marker-based augmented reality, virtual content may be precisely tracked onto pre-established photo targets. The system can accurately overlay virtual objects onto the markers while preserving alignment and perspective since the markers are carefully defined.[2]

Markerless Augmented Reality Application for Interior Designing by Nagashree R R, Rakshitha M, Ruchira Bharadwaj S, Shabana Sultana, and Narasimha Mahesh Nadig.

With markerless augmented reality, you may examine and place digital things in the actual environment without using markers like locations, images, QR codes, etc. All you need to do is use the screen on your tablet, smartphone, or AR glasses to accomplish this. Pokémon GO was one of the games that helped AR become famous. Numerous AR-enabled goods have now been created, including social networking app filters, virtual shoes, spectacles, and more. We usually have to download apps like Facebook or Instagram in order to view material. The development of web-based augmented reality has eliminated the need to download resource-intensive apps.

It does, however, ignore the possibilities of markerless augmented reality, whose augmentation doesn't require target images. This leaves a vacuum in the study and application of markerless augmented reality.

For the AR interior design app, the suggested method scored 91 percent accuracy, and for speech synthesis, it achieved 3.9 percent naturalness.

Enhanced Shopping Experience: The shopping experience is made more interactive by augmented reality.[3]

Performance Evaluation of Augmented Reality based 3D Modelling Furniture Application Tahir Ahmed T., Vijaya Shetty S., R. Samirasimha, Sushmitha Bedere J.

With the help of this program, users may get a glimpse of furniture in their actual setting and see how it would look in person. By employing this application, furniture sellers can obtain a competitive advantage in the market. Because the technology allows customers to see how furniture will look in the available space before they buy it, it also prevents customers from leaving, damage to the service provider's brand, income loss for the firm, and degradation of stakeholders' interests.

The article presents an augmented reality application for visualizing furniture, although it skips over a lot of the current augmented reality uses in the furniture sector.

The method that was suggested was able to attain a 93 percent accuracy rate for AR interior design applications and a 3.6 percent naturalness rate for speech synthesis.

Interaction: Users can move, rotate, and resize the furniture virtually, as well as rotate it within their physical area.[4]

Economical VR/AR technique for Interior Design Program, M. Izani, S. Aalkhalidi, A. Razak, and S. Ibrahim.

This study looks at how technology, specifically augmented reality (AR) and virtual reality (VR), can improve on the visualization techniques used in this field today. The potential technologies in this area were listed and explained in detail with the goal of implementing CFAD in the future. The most current technical developments in the industry serve as the foundation for this idea. The majority of students, according to the findings, showed awe at the application of VR and AR in interior design and stated a desire to learn more if it were included in the curriculum.

It is difficult for multiple individuals to collaborate on an AR interior design project at the same time since these apps lack robust collaboration features.

The method that was suggested was able to attain a 96 percent accuracy rate for AR interior design applications and a 3.5 percent naturalness rate for speech synthesis.

Remote Collaboration: Regardless of location, augmented reality (AR) enables designers, clients, and other stakeholders to virtually view and interact with design concepts in real-time. This promotes remote collaboration.[5]

Teresa, F. I. Maulana, Y. H. Cipta, A. Anggala, and I. B. A. Wijaya Kindergarten Interior Interactive Elements using Augmented Reality Design Using the ADDIE Model.

The present study used the ADDIE technique, which stands for Analysis, Design, Development, Implementation, and Evaluations, due to its perceived rationality and comprehensibility.The experiment findings show that the longer the marker tracking test, the further away the marker becomes.However,the tracking speed rises as the distance between the marker and the camera grows. It takes an average of 0.018 seconds to detect a marker. All found markers show similar 3D objects at distances of 20 to 80 cm and test tilting angles of 15 to 45 degrees. All detected markers display identical 3D objects at distances of 20 to 80 cm, and test tilt angles between 15 and 45 degrees.

It is possible that some educational institutions lack the essential resources, like dependable internet connections, appropriate hardware, and technical assistance, needed to successfully incorporate augmented reality (AR) into the classroom.

The method that was suggested was able to attain a 93 percent accuracy rate for AR interior design applications and a 3.7 percent naturalness rate for speech synthesis.

Interactive Education: Augmented Reality facilitates interactive education without requiring costly hardware or tangible resources.[6]

P. Xu Building an Augmented Reality-Based Virtual Simulation System for Interior Design.

Compared to conventional design expression techniques, the communication of design schemes is more efficient and can better meet customer requirements. The goal of this effort is to create a virtual simulation system and use augmented reality to research interior design. The simulation experiment shows that the system designed in this work has a high success rate in the test of LED light modeling situation, achieving 81.85 percentage. The average success rate of the traditional simulation system is only 47.3 percentage. When compared to the previous system, the system developed in this work has a lower average return rate—just 17.13 percentage—than the conventional simulation system.

The proposed virtual simulation system is only compared to conventional design expression techniques in this project. Its reach could be expanded by contrasting it with other cutting-edge digital design platforms or tools on the market.

For the AR interior design app, the suggested method scored 97 percent accuracy, while for speech synthesis, it achieved 3.6 percent naturalness.

Cost and Time Efficiency: Physical prototypes and several rounds of changes are common in traditional interior design procedures, which can be expensive and time-consuming.[7]

Y. Yang, S. Chae, I. Kim, Y. J. Park, and T. -d. Han Interior design-focused portable projection-based augmented reality device called DesignAR.

This study addresses the labor, resource, and time constraints associated with do-it-yourself interior design by presenting a portable augmented reality prototype system.

Any technology's practical deployment requires careful consideration of its long-term use and maintenance requirements. In order to guarantee that DesignAR remains effective, it would be beneficial to investigate how it might be updated, maintained, and supported in the future.

The method that was suggested was able to attain a 96 percent accuracy rate for AR interior design applications and a 3.5 percent naturalness rate for speech synthesis.

Flexibility and customization: Before making any long-term modifications to their actual area, users can test out various design alternatives and setups in a virtual environment thanks to AR prototyping.[8]

Enhancing the System Model for Home Interior Design with Augmented Reality by D. Kumar, P. Srinidhy, and V. P. Kafle.

In this work, we propose to design and develop a system architecture that enables users to visually try out various home interior layouts using markerless augmented reality (AR). To reduce the system latency, we employ a hybrid strategy that combines an AR framework with a Simultaneous Localization and Mapping (SLAM) method. The dynamic updating of 3D feature points prevents occlusion when numerous interiors are superimposed on the real-world view. Dimension scanning is used to precisely scale live-size content to the real world, and touch gesture recognition is used to manage transformations.

The problem is that the user's home decor may not match the desired aesthetic when it comes to shop interior settings and their actual room representation.

For the AR interior design app, the suggested method obtained 94 percent accuracy, and for speech synthesis, it achieved 3.5 percent naturalness.

Better Decision Making: The system assists users in making well-informed judgments about home décor by allowing them to virtually try out various interior arrangements.[9]

AR Development for Room Design by P. Reuksupasompon, M. Aruncharathorn, and S. Vittayakorn.

Our study offers a technique for utilizing augmented reality technology to create a furniture plan. Using the room floor plan and multiple QR markers, users can physically move furniture about the confined space (within the designated area given by the room plan) to build their own unique and stylish room arrangement. Every QR marker has an associated 3D furniture model. Finally, we demonstrate how our approach could both lessen the shortcomings of the existing room design systems and raise user satisfaction with room design applications. Moreover, we suggest that it will encourage several users to co-design concurrently.

It might be intimidating for new users of current systems because they frequently demand them to comprehend concepts like balance, scale, and proportion in interior design.

For the AR interior design app, the suggested method obtained 96 percent accuracy, and for speech synthesis, it achieved 3.8 percent naturalness.

Elegance: A space can have an air of elegance and refinement when its furniture is arranged properly.[10]

Chapter 3

System Design

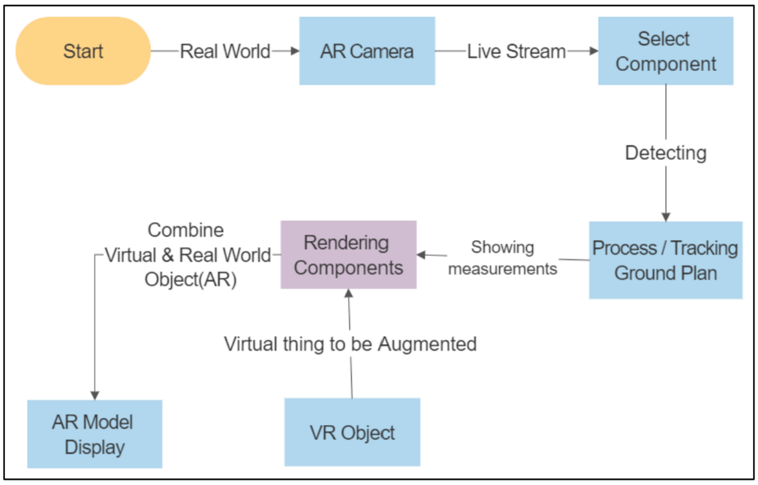

Figure 3.1: System framework figure

Because the real scene image is viewable to the human eye, the system collects photographs through a mobile built-in camera and stacks the 3D furnishings models on the screen displaying images. First and foremost, we'd want to attempt to describe the characteristics of the photos collected by the camera, select tracing images, do 3D registration, build an initial 3D map, and obtain the camera pose. Next, we create 3D furniture models with Unity 3D. The camera obtains an external parameter matrix and projection models by understanding and tracing marker images, before layering the imported 3D virtual models on the tracing image. Because Android devices consist of a touch-screen interface, we will install the furniture by dragging the screen.The system framework of this is as shown in Figure 3.1.

At the beginning of the full AR scene, monitoring mode is critical to the complete display module. This Augmented Reality system, which we organize to develop, only needs the user to interact with the mobile device and carry out no further operations in order to enjoy virtual interior augmented reality in any apprehensive indoor scenario. Initially, the environment is generated using the Ar Foundation framework SDK. That is the augmented reality camera, which will show 3D renderings in the actual world while processing and detecting the ground plane surface. A picture that can be easily accessed by any user will be boosted in the actual environment, allowing the user to feel as if it is genuine.The exhibition impacts that this approach seeks to understand include

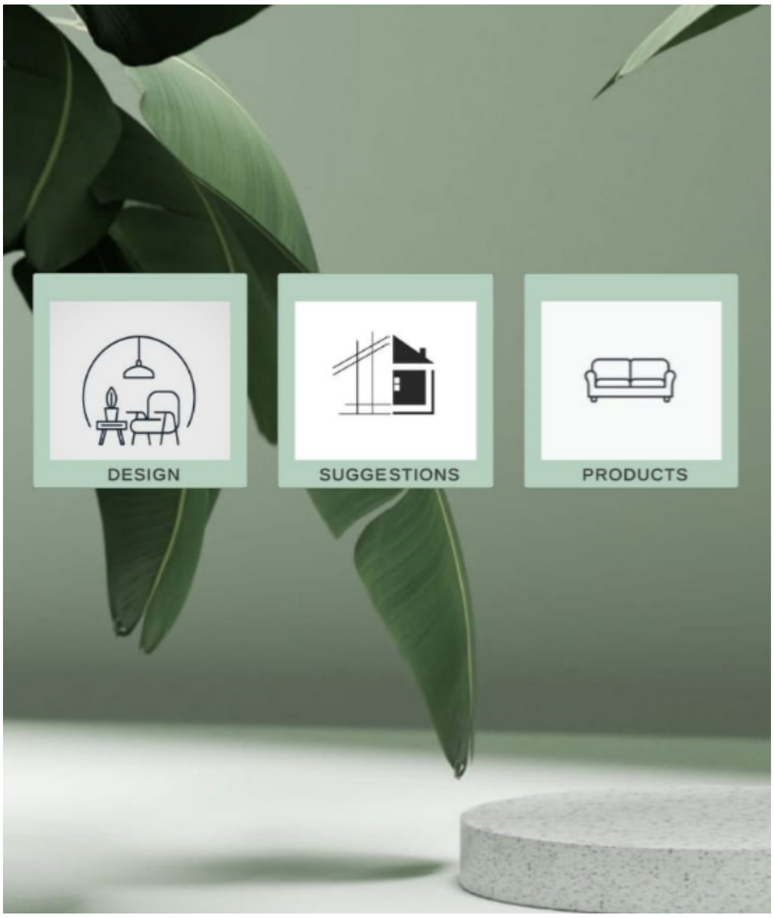

Figure 3.2: AR app

layering the virtual Component on a particular part of the space, and being able to put and view the virtual Model in real time. The display module contains the building of 3D models, the creation of 3D display visuals, and therefore the setting up of virtual component projection models, which may cause it to stack on the crucial environment. After initializing the system, we will determine the projection model from the key scene to the camera at the chosen place or region. Consistent with the Rendering approach, it then employs an AR camera to identify the position and show the virtual models in real time on the live video camera.This application is user-interactive and meant to offer a real-world viewpoint. The user can also place the 3D components where he wants. He can also get the dimensions of the 3D display elements in Ar camera, and he can zoom in and out as he desires to customize the room. He can also see all of the components at once, as well as all of the designing views in the real world, using this application.

Table 3.1: Framework used

Andriod	ARcore
IOS	ARKit

3.1 Methodology

The initial stage in this research was to develop a straightforward application for consumers to use AR. Using the Unity software, we created distinct AR application for mobile phones.When the user clicks on the app an icon with the loading message appears which transits to main menu consisting three buttons namely 'design', 'suggestions', 'products' etc. When the user clicks on the design button a camera message pops up with 2 options yes or no, when the user clicks on yes the app allows him to go to the next scene and if he clicks no the app shuts down itself. If clicked yes the app transits to a decor option scene. It consists of 3 buttons namely 'furniture', 'wallpaper', flooring'. When the furniture button is clicked it opens up to a new scene consisting 4 buttons namely 'living room'-consisting of the furnitures which are used to decorate the living room like sofa, coffee table etc, 'kitchen'- consisting of the furnitures which are used to decorate the kitchen space like dinning table, chairs etc, 'bedroom'- consisting of the furnitures which are used to decorate the bedroom like bed, wardrobe etc, 'bathroom'- consisting of the furnitures which are used to decorate the bathroom like toiletries etc. When wallpaper is clicked it opens up a new scene with different color palettes from which the user can choose to paint his space. When flooring is clicked it opens up a new scene with different flooring materials from which the user can choose to place in his space .When the user clicks on the suggestions button it opens a 3D model of a well furnished home. When the user clicks on the products button it opens up and shows different products which are used in the application.

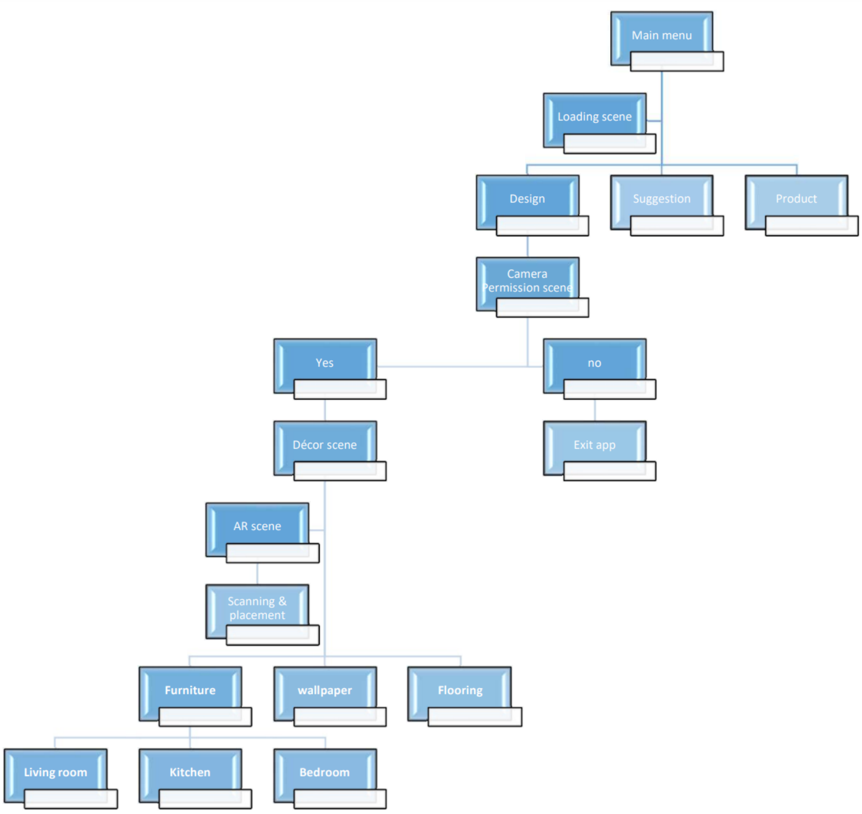

Figure 3.3: Flow Chart

Chapter 4

System Implementation

4.1 Unity Hub

We download the Unity Hub from the chrome then install an editor of our choice, we used the the 2021.3.37f1 editor for our project. Once installation is done,we create a new project and named it AR interior as shown in the Figure 4.1 and used 3D core for it because it uses Unity's build-in render.Then the unity editor opens and we start bulding our project one by one.

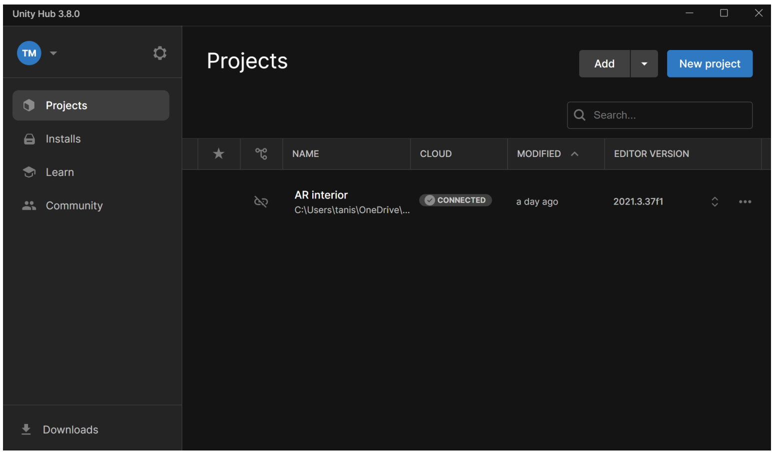

Figure 4.1: Unity Hub

4.2 Blender

To build the models we used the blender software.Blender is an open-source, feature-rich 3D creation suite that may be used for a multitude of tasks, including rendering, compositing, animation, simulation, video editing, and 3D modeling.As shown in fig 4.2.

4.3 3D model scene

In Unity, a 3D model scene is the virtual setting in which 3D objects, cameras, lighting, and other components are positioned, adjusted, and produced for interactive applications

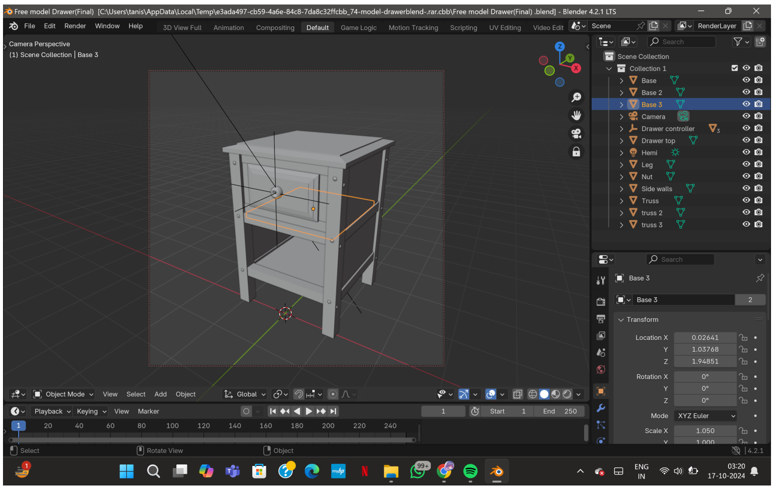

Figure 4.2: Blender

or games. Real-time tools for creating, editing, and rendering these 3D scenarios are offered by the well-known gaming engine Unity.As shown in fig 4.3.

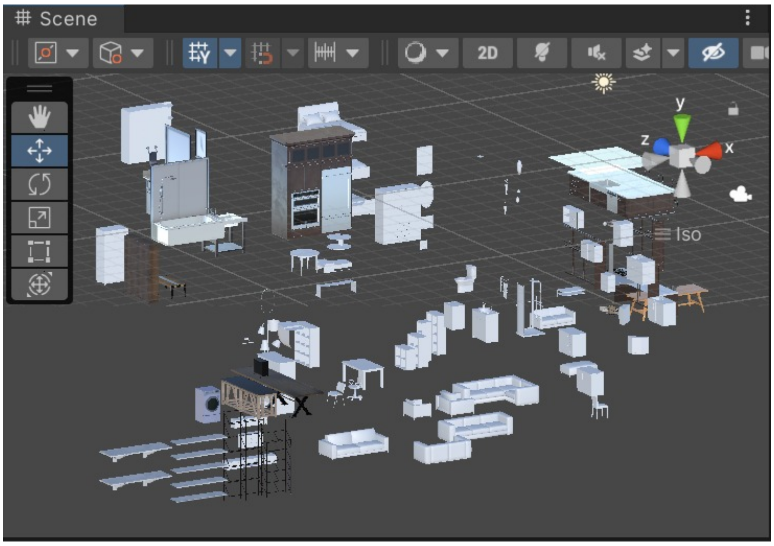

Figure 4.3: 3D model scene

4.4 Furniture's

Following are some furniture's we have built.From basic coffee table(fig 4.5) and Shelf(fig 4.4) to sophisticated bed(fig 4.7) and sofa(fig 4.6) with elaborate textures.

4.5 Loading page

An AR interior design software's loading page is the user's initial engagement with the program, and its layout is essential to ensuring a seamless experience as the app gets ready to launch.As shown in fig 4.8.

Figure 4.4: Coffee Table

Figure 4.5: Shelf

Figure 4.6: Bed

Figure 4.7: Sofa

Figure 4.8: Loading page

4.6 Decor options

This page will let you choose among three buttons i.e furniture,wallpaper and flooring.In furniture we will find several options of furnishes and in wallpaper and flooring we can choose among the best colors and textures suited for a particular space.

Figure 4.9: Decor Options

4.7 Room options

This page will let you choose among four buttons i.e Living room,Kitchen,bedroom and bathroom.

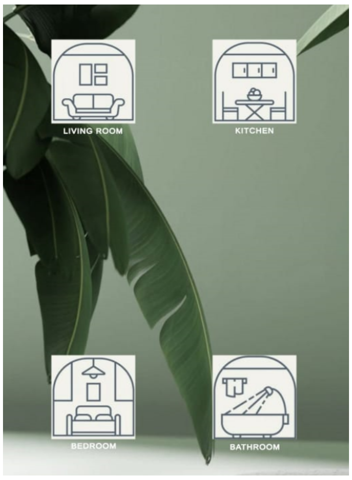

Figure 4.10: Room Options

4.8 Camera permission

The next scene will lead to asking for camera permission. If we click on yes it will lead to the next scene but if we click on no we will exit from the app.

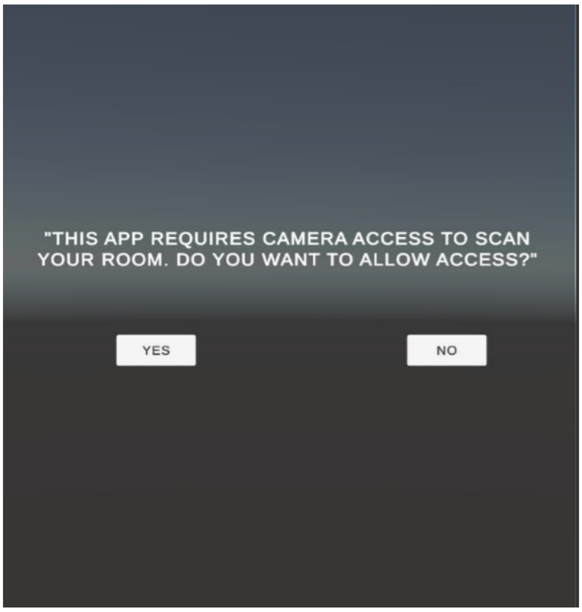

Figure 4.11: Camera permission

4.9 Script(c-sharp)

4.9.1 Main menu

It defines a class MainMenuManager that handles button click events for navigating between different scenes in a Unity application.

```
using UnityEngine;
using UnityEngine.SceneManagement;

0 references
public class MainMenuManager : MonoBehaviour
{
    // Function to handle the Design button click
    0 references
    public void OnDesignButtonClicked()
    {
        // Load the Camera Permission Scene
        SceneManager.LoadScene("CameraPermissionScene"); // Replace with the correct scene name
    }

    // Function to handle the Suggestions button click
    0 references
    public void OnSuggestionsButtonClicked()
    {
        // Load the Suggestions scene (or another scene that shows suggestions)
        SceneManager.LoadScene("SuggestionsScene"); // Replace with the correct scene name
    }

    // Function to handle the Products button click
    0 references
    public void OnProductsButtonClicked()
    {
        // Load the Products scene
        SceneManager.LoadScene("ProductsScene"); // Replace with the correct scene name
    }
}
```

Figure 4.12: Main menu

4.9.2 Decor management

This class is responsible for handling button clicks that navigate the user to different scenes within the application, like for furniture, wallpaper, and flooring.

```
Assets > Scripts > DecorManager.cs > DecorManager
using UnityEngine;
using UnityEngine.SceneManagement;

0 references
public class DecorManager : MonoBehaviour
{
    // This function loads the Furniture scene
    0 references
    public void OnFurnitureButtonClicked()
    {
        SceneManager.LoadScene("FurnitureScene"); // Replace with the exact name of your Furniture scene
    }

    // This function loads the Wallpaper scene
    0 references
    public void OnWallpaperButtonClicked()
    {
        SceneManager.LoadScene("WallpaperScene"); // Replace with the exact name of your Wallpaper scene
    }

    // This function loads the Flooring scene
    0 references
    public void OnFlooringButtonClicked()
    {
        SceneManager.LoadScene("FlooringScene"); // Replace with the exact name of your Flooring scene
    }
}
```

Figure 4.13: Decor Management

4.9.3 Wallpaper Changer

It defines a class WallpaperChnager that allows the user to choose among various wallpaper options.

```
using UnityEngine;
using UnityEngine.UI;

0 references
public class WallpaperChanger : MonoBehaviour
{
    1 reference
    public Renderer wallRenderer;
    2 references
    public Material[] wallpaperMaterials;
    2 references
    public Button[] wallpaperButtons;

    0 references
    void Start()
    {
        // Assign button click events
        for (int i = 0; i < wallpaperButtons.Length; i++)
        {
            int index = i;  // Important to capture index in a separate variable
            wallpaperButtons[i].onClick.AddListener(() => ChangeWallpaper(index));
        }
    }

    1 reference
    public void ChangeWallpaper(int index)
    {
        // Change the material of the wall
        if(index >= 0 && index < wallpaperMaterials.Length)
        {
```

Figure 4.14: Wallpaper changer

4.10 Suggestion Rooms

Following are the rooms we have created, recommending the users our own designs that might fit their choice and preference for their house decor.

Figure 4.15: Bedroom

Figure 4.16: Kitchen

Figure 4.17: Bathroom

Figure 4.18: Hallway

4.11 Flooring options

The following image shows the flooring options we have provided to the users, which they can choose and access according to their preferences.

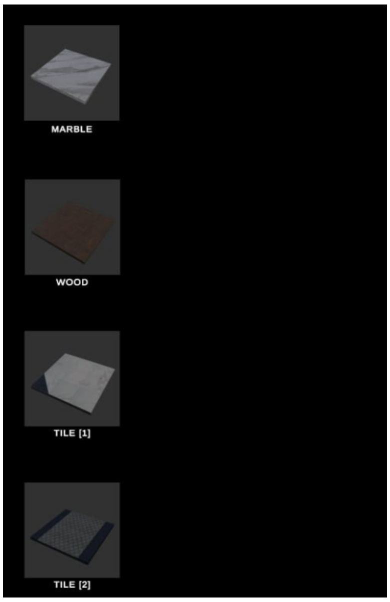

Figure 4.19: Flooring options

4.12 Sofa Placing in real world

This is the output we get from building the project. So when we open our AR camera and direct it on a surface it will scan the plane and place our selected object(furnitures).

Figure 4.20: Sofa Placing in real world

Chapter 5

Result and Analysis

By enabling customers to see how furniture and décor will appear in their real living environments before making purchases, integrating Augmented Reality (AR) into an interior design application greatly improves the user experience. People spend more time playing with various designs and materials in real-time thanks to this feature, which boosts user engagement and promotes social sharing and organic reach. Because users are more confident in their design choices and are less likely to return items, augmented reality (AR) also increases conversion rates.

Chapter 6

Conclusion and Future scope

6.0.1 Conclusion

In this article, we suggest an augmented reality (AR) application for room design that aims to let consumers see furniture arrangements in a virtual area before purchasing the furnishings.The proposed approach uses augmented reality (AR) technology to identify and detect the position of a marker or QR code in a live video stream.

6.0.2 Future Scope

The future scope of AR interior design apps is highly promising as advancements in technology continue to enhance user experiences and capabilities.

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